



Customer Shopping Behavior Analysis

Uncovering insights from **3,900 transactions** across product categories — spending patterns, customer segments, and subscription behavior to guide strategic decisions.

Dataset at a Glance

3,900

Total Purchases

Transactional records analyzed

18

Features

Demographics, purchases & behavior

\$59.76

Avg. Purchase

Average spend per transaction

37

Missing Values

In Review Rating column only

Key features include customer demographics (age, gender, location, subscription status), purchase details (item, category, amount, season), and shopping behavior (discounts, frequency, review ratings, shipping type).

Data Preparation in Python

Cleaning & Standardization

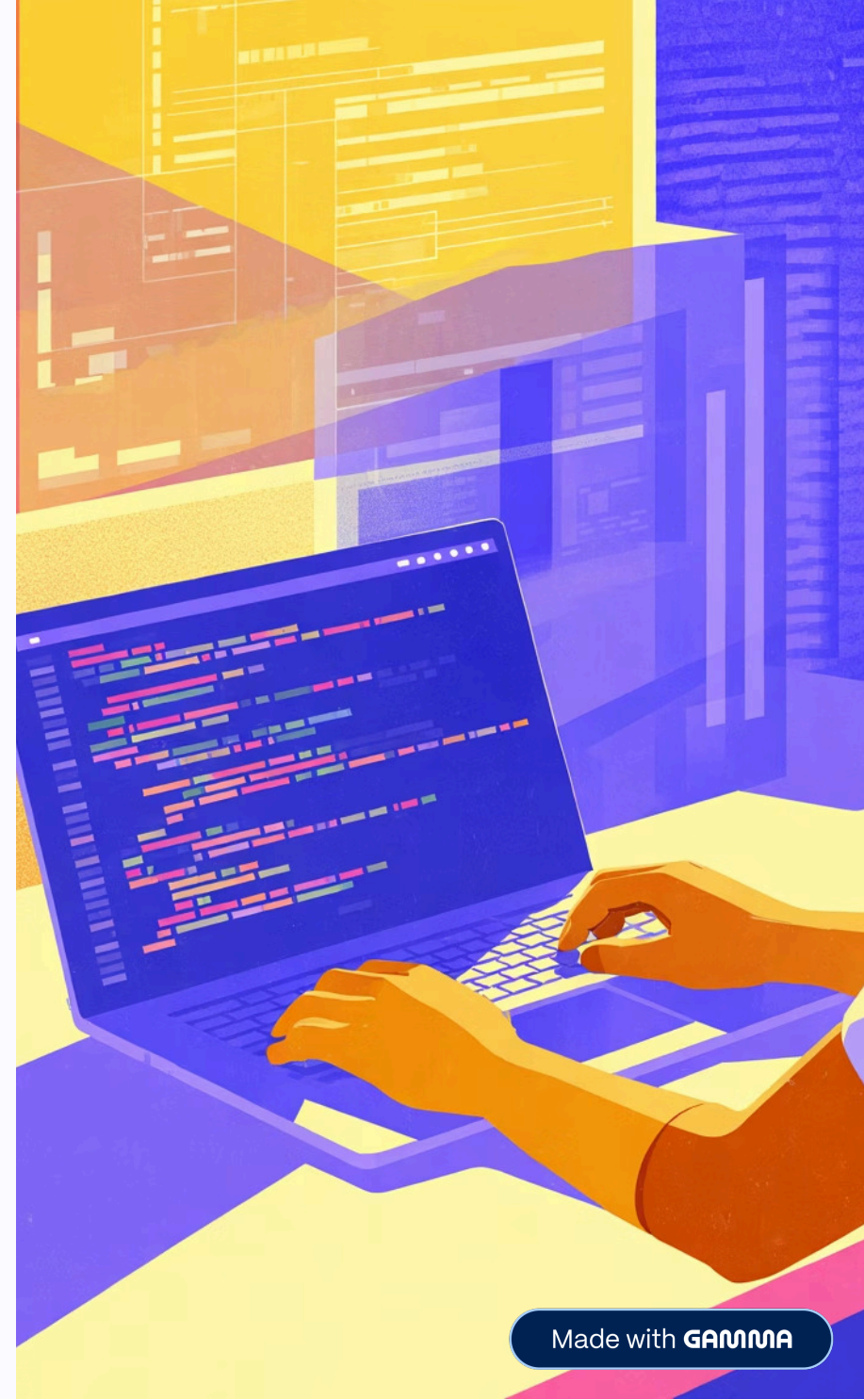
Loaded data via `pandas`, checked structure with `df.info()` and `.describe()`, imputed missing Review Ratings using median by category, and renamed columns to snake_case.

Feature Engineering

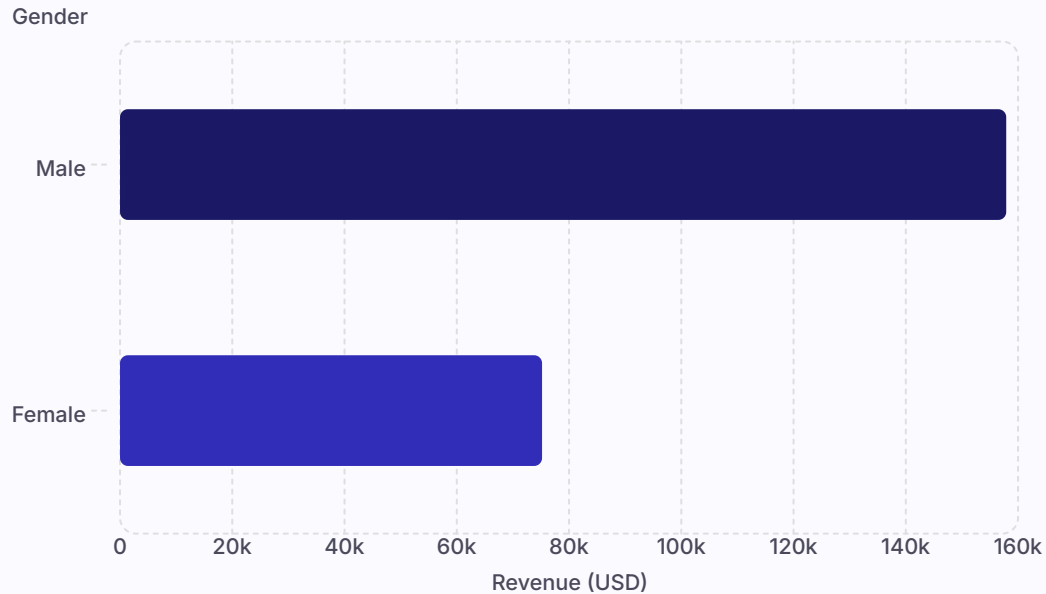
Created `age_group` by binning customer ages and `purchase_frequency_days` from purchase data. Dropped redundant `promo_code_used` column.

Database Integration

Connected Python to PostgreSQL and loaded the cleaned DataFrame for SQL analysis.



Revenue by Gender



Key Insight

Male customers generated **more than double** the revenue of female customers — **\$157,890 vs. \$75,191** — representing a significant opportunity to grow female customer engagement and spending.

Top Products & Discount Patterns

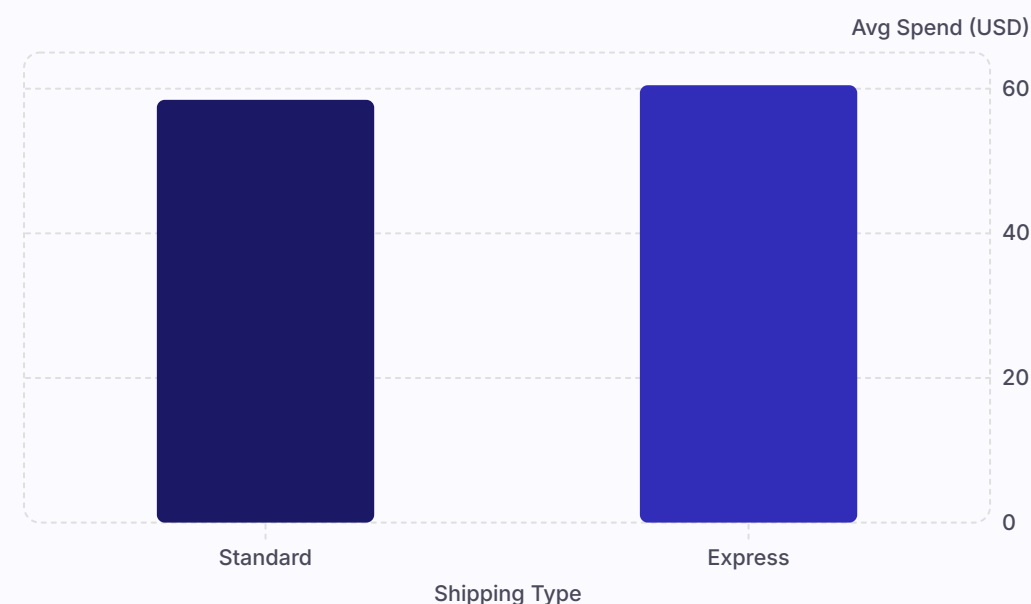
Top 5 Products by Rating

Product	Avg Rating
Gloves	3.86
Sandals	3.84
Boots	3.82
Hat	3.80
Skirt	3.78

Most Discount-Dependent Products

Product	Discount Rate
Hat	50.00%
Sneakers	49.66%
Coat	49.07%
Sweater	48.17%
Pants	47.37%

Shipping & Subscription Insights

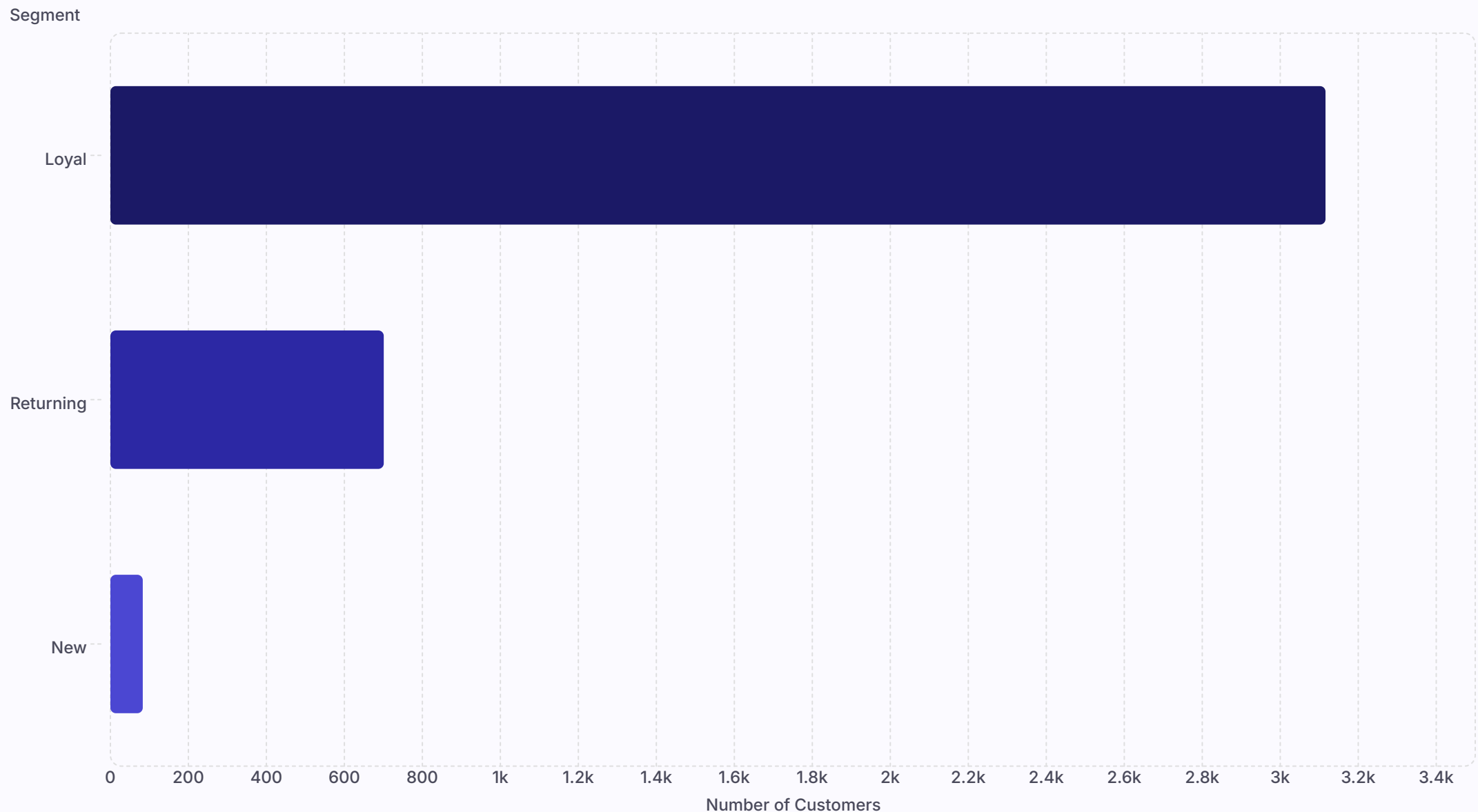


Subscribers vs. Non-Subscribers

Status	Customers	Revenue
Subscribed	1,053	\$62,645
Not Subscribed	2,847	\$170,436

Express shipping customers spend slightly more (\$60.48 vs. \$58.46). Non-subscribers vastly outnumber subscribers (**73% vs. 27%**) — a major conversion opportunity.

Customer Segmentation



Customers were classified into three segments based on purchase history. **Loyal customers** dominate at 3,116 — representing the core revenue base. Of 3,476 repeat buyers (more than 5 purchases), 958 are subscribers while 2,518 are not, indicating strong potential for subscription conversion among loyal shoppers.

Top Products by Category

Accessories

Jewelry · Sunglasses · Belt

Clothing

Blouse · Pants · Shirt

Footwear

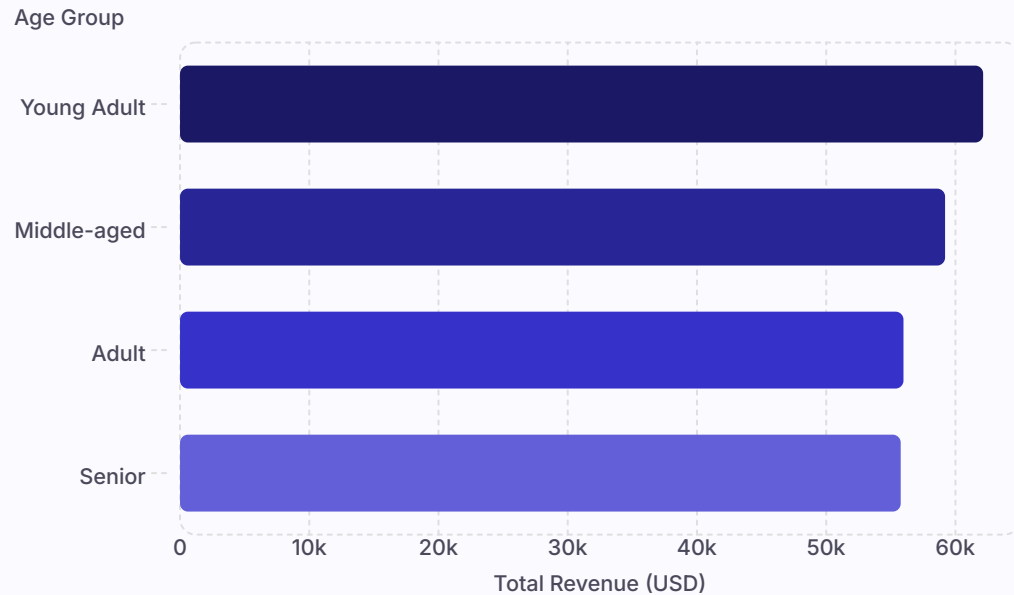
Sandals · Shoes · Sneakers

Outerwear

Jacket · Coat



Revenue by Age Group



Age Group Breakdown

Revenue is relatively balanced across age groups. **Young Adults** lead with \$62,143, while Seniors contribute the least at \$55,763 — a narrow \$6,380 spread across all segments, suggesting broad cross-generational appeal.

Power BI Dashboard & Recommendations

An interactive Power BI dashboard presents KPIs (3.9K customers, \$59.76 avg purchase, 3.75 avg rating) with filters for subscription, gender, category, and shipping.



Boost Subscriptions

Promote exclusive subscriber benefits to convert the 73% non-subscribed base.



Loyalty Programs

Reward repeat buyers to grow the Loyal segment.



Review Discount Policy

Balance sales boosts with margin control on discount-heavy products.



Targeted Marketing

Focus on high-revenue age groups and express-shipping users.



Product Positioning

Highlight top-rated and best-selling products in campaigns.